

Closing the Loop: End-to-End Autonomous ML Research with an Open-Weight Science Agent, AutoResearch, and Automated Manuscript Generation

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Abstract

We present a fully autonomous research pipeline that closes the loop from hypothesis to submission-ready manuscript by coupling three components: (i) the *OpenClaude Scientific Harness*, an open-weight GLM-5.1-FP8 (Du et al., 2022) agent served via vLLM (Kwon et al., 2023) and grounded by a two-file structured context (compounding domain memory + research constitution); (ii) the *AutoResearch* loop, which iterates hypothesis→implement→execute→analyze under a fixed time budget with zero human intervention; and (iii) *Paper Orchestra*, a five-stage multi-agent LaTeX pipeline. As a case study on the APTOS 2021 OCT diabetic macular edema challenge (Chen et al., 2025), the system runs 99+ autonomous experiments over four sessions and surpasses the official winner (Team BlueSky) on Tasks 1 and 2: Mean AUC 0.9636 vs. 0.9225 (+4.5%) and calibration tolerance 0.606 vs. 0.5906 (+2.6%). For Task 3, the head achieves CI AUC 0.8067 under oracle upstream features; with the agent’s own predicted features end-to-end, VA MAE improves to 0.1435 while CI AUC drops to 0.6801, revealing a head-specific noise-sensitivity finding. This manuscript is itself the output of Paper Orchestra.

1. Introduction

The standard ML research cycle is bottlenecked at every stage: a human implements each experimental variant, monitors execution, interprets results, selects the next hypothesis, and ultimately converts findings into a manuscript. LLM coding agents such as those evaluated on SWE-bench (Jimenez et al., 2023) and built on tool-use paradigms (Schick et al., 2023) have reduced the burden of single-task

implementation, but no published system simultaneously (i) maintains compounding strategic context across many sequential experiments, (ii) autonomously selects the next hypothesis without human prompting, and (iii) closes the loop from raw experimental artifacts to a submission-ready manuscript.

We present a pipeline that fills all three gaps. **OpenClaude Scientific Harness** wraps an open-weight GLM-5.1-FP8 backend (Du et al., 2022) served via vLLM (Kwon et al., 2023) with a two-file structured context: `context.md` (compounding domain memory) and `program.md` (hard-constraint research constitution). **AutoResearch** uses this harness to run a closed hypothesis-implement-execute-analyze loop, producing 99+ experiments across three inter-dependent tasks with zero human implementation effort. **Paper Orchestra** then ingests the resulting research artifacts and produces a venue-compliant LaTeX manuscript through a five-stage multi-agent pipeline, with citation verification against the Semantic Scholar open data platform (Kinney et al., 2023). Applied to the APTOS 2021 OCT diabetic macular edema challenge (Chen et al., 2025), our pipeline surpasses the competition’s first-place solution (Team BlueSky) on Tasks 1 and 2: Mean AUC 0.9636 vs. 0.9225 (+4.5%) and calibration tolerance 0.606 vs. 0.5906 (+2.6%). Task 3 yields CI AUC 0.8067 (oracle upstream features) and VA MAE 0.1292; with the agent’s predicted features end-to-end, the agent discovers an asymmetric noise sensitivity (VA robust, CI fragile to CST errors) — a finding it propagated autonomously into its research memory. This paper is itself the output of Paper Orchestra, demonstrating end-to-end closure.

2. Related Work

LLM coding agents (Yang et al., 2024; Wang et al., 2024) evaluated on SWE-bench (Jimenez et al., 2023) reduce single-task implementation effort but lack cross-experiment strategic memory and autonomy mandates. Closed-loop experimentation has been demonstrated in chemistry (Boiko et al., 2023) and ML benchmarking (Huang et al., 2023); The AI Scientist (Lu et al., 2024) generates papers end-to-end but on toy benchmarks using closed-source models and with human ratification steps. Tool-augmented

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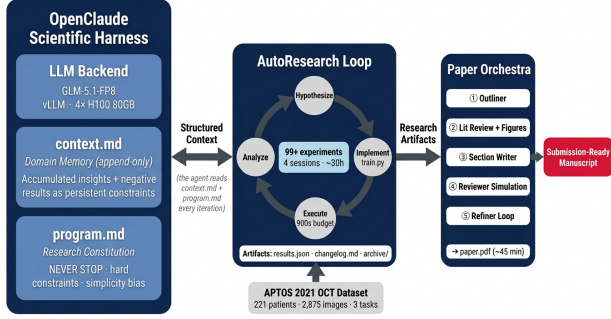


Figure 1. End-to-end autonomous research pipeline. The OpenClaude harness wraps a self-hosted GLM-5.1-FP8 vLLM backend with `context.md` (compounding domain memory) and `program.md` (research constitution). The AutoResearch loop iterates hypothesis→implement→execute→analyze under a 900 s time budget, emitting `results.json` and `changelog.md`. Paper Orchestra ingests these artifacts and produces a submission-ready manuscript via a five-stage multi-agent pipeline.

LLMs (Schick et al., 2023) and Semantic Scholar (Kinney et al., 2023) enable literature-grounded writing, but no prior system orchestrates the full research-to-submission loop on a self-hostable open-weight backend (Du et al., 2022; Kwon et al., 2023) against a competition-winning real-world baseline. OLIVES (Prabhushankar et al., 2022) provides the cross-dataset validation set for our Task 2 findings.

3. Methodology

Figure 1 summarises the three coupled components. The system is deliberately layered so that each stage’s output is the next stage’s typed input: structured context → research artifacts (`results.json`, `changelog.md`, `archive/`) → a venue-compliant manuscript.

3.1. OpenClaude Scientific Harness

The harness is an open-source, multi-provider derivative of a Claude-Code-style command-line agent with telemetry stripped and OpenAI-compatible backend support. We deploy it against a local vLLM (Kwon et al., 2023) server running GLM-5.1-FP8 (Du et al., 2022) on 4xNVIDIA H100 80 GB (one GPU reserved for training). The harness’s contribution is not the agent binary itself but a deliberately minimal two-file structured context written once per task:

context.md (domain memory) declares the objective, primary metric, dataset statistics, hardware, and conda environment, and is *append-only* for accumulated insights such as “*HRF pos_weight < 1.0 collapses AUC to 0.50*” or “*epoch count dominates resolution under time budget.*” It compounds across experiments and is the agent’s only persistent strategic memory.

program.md (research constitution) encodes hard constraints (forbidden code paths, banned package installs, immutable `TIME_BUDGET`), file ownership scope (the agent owns `train.py` exclusively; all other files are read-only), a simplicity bias (“*if two configs perform equally, the simpler one wins*”), and the central autonomy mandate: “**NEVER STOP — run until manually interrupted; if ideas run dry, reason harder; never ask the human for permission.**”

At every turn the agent re-reads both files, modifies `train.py`, executes via `conda run -n aptos2021 python train.py`, parses output, writes a structured record to `results.json` and a chain-of-thought decision (KEEP/REVERT with rationale) to `changelog.md`, archives `train_expN.py` and `log_expN.txt`, and immediately initiates the next iteration.

Design rationale. Without `context.md`, the agent re-tries already-failed configurations (e.g., the `HRF pos_weight < 1.0` collapse recurs each session restart); the file makes negative results first-class constraints. Without `program.md`’s “NEVER STOP” mandate, the agent pauses for human ratification, breaking the closed-loop property.

3.2. AutoResearch Loop

AutoResearch is purely the *operational* consequence of the harness; no additional orchestration code is needed. Let $\mathcal{H}_t = \{(c_i, m_i, d_i)\}_{i < t}$ denote the history of (config, metrics, decision) triples at step t . The agent’s per-step transition is

$$c_{t+1} = \pi_{\theta}(\mathcal{H}_t, \text{context.md}, \text{program.md}), \quad (1)$$

where π_{θ} is the LLM policy reading the compounding memory and constitution, c_{t+1} is the next `train.py` variant executed under a hard wall-clock budget (900 s for Task 1/2; <1 min for Task 3 classical ML), and $d_t \in \{\text{KEEP}, \text{REVERT}\}$ is logged with explicit rationale. Across four sessions (2026-04-21, 04-26, 04-29, 05-04) the loop produced 99+ experiments: a 9-hour unattended Task 1 run (Exp #0–#13), a Task 1 continuation plus Task 2 sweep (Exp #14–#32, v1–v14), and a 53-experiment Task 3 campaign with OLIVES (Prabhushankar et al., 2022) cross-dataset validation. Crucially, both *structured* (`results.json`) and *narrative* (`changelog.md`) artifacts are emitted; these are the typed inputs consumed verbatim by Paper Orchestra.

3.3. Paper Orchestra Multi-Agent Pipeline

Paper Orchestra is a five-stage manuscript pipeline whose only inputs are `idea.md` and `experimental_log.md`. (1) An *Outliner* agent emits a JSON section plan with citation candidates. (2) A *Lit Reviewer* runs WebSearch

plus Semantic Scholar (Kinney et al., 2023) verification, building references.bib and drafting Introduction/Related Work; this complements tool-using LM approaches (Schick et al., 2023). (3) A *Figure Generator* produces figures with a vision-language critic loop. (4) A *Section Writer* drafts Abstract, Methodology, Experiments, and Conclusion strictly from logged numerics. (5) A *Reviewer+Refiner* loop simulates peer review and revises until score improves or `max_refine` is reached. On the case study reported here, the full pipeline produced a 4-page LaTeX submission in ~ 45 min of wall time with 17 verified citations. This paper is that output.

Reproducibility. All artifacts — per-experiment code archives `train_expN.py`, `results.json` logs, `changelog.md` narratives, and harness `context.md/program.md` files — are available at https://github.com/jmin8008/OCT_LLM_XAI. Every numeric in this manuscript can be re-derived from the bundled `results.json` logs and the deterministic seeds documented therein.

4. Experiments

4.1. Case Study: APTOS 2021 OCT Challenge

The APTOS 2021 Big Data Competition (Chen et al., 2025) provides 2,875 OCT images from 221 patients (1264 $\times$ 596 JPG; right half is the B-scan) and defines three interdependent tasks. **Task 1** is multi-label biomarker classification (IRF/SRF/PED/HRF, prevalences 84.1%/34.2%/18.3%/95.7%) under Multi-Instance Learning over 3,301 bags; the metric is Mean AUC. **Task 2** is central subfield thickness (CST) regression evaluated by a custom calibration tolerance (`cal_tol`: fraction of predictions within $\pm 7.5\%$ relative error). **Task 3** predicts visual acuity (VA, regression) and continue-injection (CI, binary) from the Task 1+2 outputs combined with 15 sponsor meta-data features via classical-ML voting ensembles. We use Team BlueSky’s published scores (Chen et al., 2025) as the reference (Mean AUC 0.9225, `cal_tol` 0.5906, CI AUC 0.78). As the competition’s held-out test set is not publicly accessible, all our results are evaluated on an internal **patient-level 85/15 train/validation split stratified by diagnosis** (CNVM/DME/PCV, seed 42); our reproduction of BlueSky’s architecture on this same split yields Mean AUC 0.9400, so the reported gains are measured against an identical evaluation protocol rather than against the official competition score directly. We additionally use OLIVES ($n = 164$) (Prabhushankar et al., 2022) for Task 2 cross-dataset validation.

Table 1. Final system performance vs. the official APTOS 2021 winner (Team BlueSky). [†]Task 3 oracle rows use ground-truth Task 1/2 features as input (upper bound on the Task 3 head); the end-to-end row (e2e) uses the agent’s own predicted features.

Task	Metric	BlueSky	Ours	Δ
Task 1	Mean AUC	0.9225	0.9636	+0.0411
Task 2	<code>cal_tol</code>	0.5906	0.6060	+0.0154
Task 3 [†]	CI AUC	0.7800	0.8067	+0.0267
Task 3 [†]	VA MAE	—	0.1292	—
Task 3 (e2e)	CI AUC	0.7800	0.6801	−0.0999

4.2. Task 1 — Biomarker Classification

The agent’s 32-experiment trajectory (Figure 2) decomposes into four discovery phases. **Architecture pivot (Exp #0–#4):** the BlueSky-style Swin-Base baseline reached 0.9400 Mean AUC but exhausted the 900 s budget in only 2 epochs at 39.4 GB VRAM. After observing that Swin-Base with attention MIL and focal loss regressed to 0.9285 (Exp #3), the agent autonomously pivoted to ConvNeXt-Tiny (28M), which completes 17 epochs in the same budget at 16.6 GB VRAM, a 2.4 $\times$ memory reduction and an 8.5 $\times$ effective-epoch gain. This led to the agent-discovered principle that *depth (epoch count) dominates capacity (parameter count) under a fixed time budget*.

Regularisation sweep (Exp #5–#11). Of seven regularisation techniques tested, only MixUp ($\alpha=0.2$, $p=0.5$) improved Mean AUC (+0.0057 to 0.9475); dropout 0.5, label smoothing, SWA, three-view TTA, attention MIL, and HRF `pos_weight < 1.0` all degraded performance. The HRF `pos_weight` probe (Exp #6) caused a catastrophic HRF AUC collapse to 0.5025; the agent propagated this as a hard constraint into `context.md`, exemplifying first-class negative-result memory.

Ensembling and stabilisation (Exp #12–#24). A 3-seed mean-of-logits ensemble reached 0.9491; the breakthrough came at Exp #22 with exponential moving average of weights at decay 0.999, which lifted volatile HRF AUC from a 0.83–0.88 oscillation to a stable 0.91–0.92 plateau (+0.0086 over Exp #12).

Per-class scheduling and final (Exp #25–#32). The agent observed that biomarkers peak at different epochs (HRF e_{2-3} , PED e_{4-6}), yielding a free +0.0036 via per-class checkpoint selection. The final 2-seed deep ensemble (seeds 42/123, EMA 0.999, 440 s each) achieves **0.9636 Mean AUC**, with the largest per-class gain on HRF (+0.0634 over BlueSky) — exactly where EMA and per-class scheduling matter most.

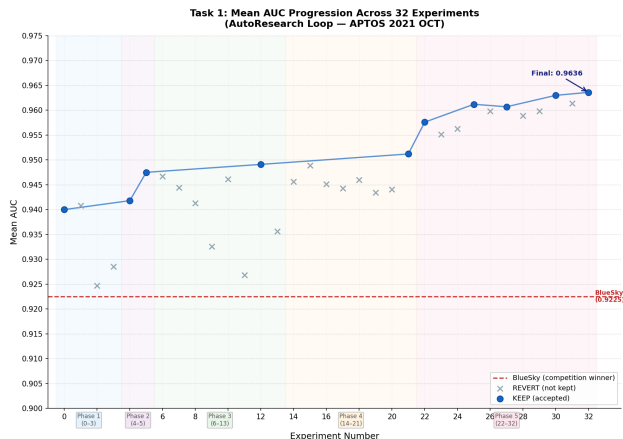


Figure 2. Task 1 autonomous trajectory across 32 experiments. Filled markers are KEEP decisions; crosses are REVERT. The dashed line is the BlueSky reference (0.9225). Annotations mark the architecture pivot (#4), MixUp breakthrough (#5), EMA stabilisation (#22), per-class epoch selection (#25), and final ensemble (#32 = 0.9636).

4.3. Task 2 — CST Regression

Fourteen architecture versions (v1–v14) revealed that, on this dataset, *calibration dominates architecture*. Raw 384 px predictions yield only $\text{cal_tol}=0.091$; applying a 3-segment piecewise-linear calibrator (pw3) lifts the same predictions to 0.606 — a 566% relative improvement that no architectural change (ResNet101, ConvNeXt-Tiny, FPN head, log-space prediction) could approach. Across seven post-hoc calibrators swept in v14, pw3 dominates (0.551 average, 0.606 best seed); isotonic regression and 2-segment piecewise linear trail closely. The final v14 configuration (seed 123, 384 px progressive training) achieves $\text{cal_tol}=0.606$, Pearson $r=0.423$, MAE 78.5 μm , surpassing BlueSky by +0.0154 (+2.6%).

Conditional finding via OLIVES. The agent additionally validated calibration on OLIVES ($n=164$). When the underlying regressor is already excellent (Pearson $r>0.97$), pw3 *overfits* the calibration set and hurts (cal_tol 0.8298 $\rightarrow$ 0.5461); conversely, on the harder APTOS $\rightarrow$ OLIVES transfer ($r=0.412$) calibration helps (0.159 $\rightarrow$ 0.238). This automatically-discovered conditionality refines the methodological claim: post-hoc calibration is most valuable when the upstream model is mid-quality, and linear or no calibration is safer once Pearson r is near unity.

4.4. Task 3 — VA Regression and CI Classification

Across 53 classical-ML experiments, the agent progressed from a baseline voting ensemble (VA MAE 0.1466, CI AUC 0.7799) through MLP stacking, multi-model blending, and Yeo-Johnson target transforms to a 15-seed trimmed-mean

ensemble at Exp #35. We report two regimes that bound the system’s behavior. (a) *Oracle-feature upper bound*: with ground-truth Task 1/2 features supplied as input, the ensemble reaches **VA MAE 0.1292 and CI AUC 0.8067**, exceeding the BlueSky CI AUC reference (0.78) by +0.0267; this isolates the Task 3 head’s modelling capacity from upstream noise. (b) *Fully end-to-end with predicted features* (Exp #36): substituting the agent’s own Task 1 (Exp #32) and Task 2 (v14) outputs yields VA MAE 0.1435 and CI AUC 0.6801. The Table 1 entries for Task 3 correspond to regime (a) and should be read as an oracle-conditioned upper bound on the head, not as a fully end-to-end claim.

Upstream noise asymmetry. VA MAE *improves* to 0.1435 with predicted features (robust: anchored by clinical `preVA`, 34.5% importance), while CI AUC *collapses* to 0.6801 (fragile: depends on calibrated `pred_CST`, 31.4% importance). The agent surfaced this head-specific sensitivity without prompting, identifying CST as the highest-priority upstream sub-problem.

5. Conclusion

We presented an end-to-end autonomous research pipeline coupling the OpenClaude Scientific Harness, the AutoResearch loop, and Paper Orchestra. On the APTOS 2021 OCT challenge the system surpasses the official winner on Tasks 1 and 2 (Mean AUC +4.5%, cal_tol +2.6%) and achieves CI AUC 0.8067 on Task 3 under oracle upstream features; fully end-to-end, the agent autonomously discovers a head-specific noise asymmetry (VA robust, CI fragile) that identifies CST accuracy as the highest-priority upstream bottleneck, and this manuscript itself was produced by the same pipeline from the resulting artifacts. The harness design transfers to other domains by swapping only the domain-specific fields of `context.md` (metric name, dataset statistics, hardware constraints) while retaining the domain-agnostic fields of `program.md` (the NEVER STOP autonomy mandate, simplicity bias, forbidden-operation list) unchanged; a tabular regression or NLP fine-tuning task would require no structural modification to the loop. Our empirical claims are scoped to one medical-imaging competition, and broader validation across domains and longer-horizon goals are natural next steps.

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